Exercise 1

Let $Z \sim \mathcal{N}(0,1)$. Prove that

$$\mathbb{E}\left[e^{aZ-\frac{1}{2}a^2}\mathbf{1}_{\{Z>b\}}\right] = N(a-b).$$

Solution

By definition of the expected value and pdf of a normal distribution, we obtain

$$\begin{aligned}
\mathbb{E}\left[e^{aZ-\frac{1}{2}a^2}\mathbf{1}_{\{Z>b\}}\right] &= \int_b^\infty e^{az-\frac{1}{2}a^2}\frac{1}{\sqrt{2\pi}}e^{-z^2/2}\,dz \\
&= \frac{1}{\sqrt{2\pi}}\int_b^\infty e^{-\frac{1}{2}\left(z^2-2az+a^2\right)}\,dz \\
&= \frac{1}{\sqrt{2\pi}}\int_b^\infty e^{-\frac{1}{2}(z-a)^2}\,dz.
\end{aligned}$$

Using the substitution

$$u = z-a, \qquad dz = du,$$

we get

$$\begin{aligned}
\mathbb{E}\left[e^{aZ-\frac{1}{2}a^2}\mathbf{1}_{\{Z>b\}}\right] &= \frac{1}{\sqrt{2\pi}}\int_{b-a}^\infty e^{-u^2/2}\,du. \\
&= 1-N(b-a) \\
&= N(b-a),
\end{aligned}$$

where the last equality holds by the symmetry of the standard normal distribution.

Exercise 2

Let C_t and P_t denote respectively the prices at time t of a European call and put option with strike K and maturity T on an underlying asset with price S_t. Prove the put-call parity for $t<T$:

$$C_t - P_t = S_t - Ke^{-r(T-t)}.$$

Proof

Consider the following two portfolios at time t.

- Portfolio A contains one European call option and the cash amount $Ke^{-r(T-t)}$. Its value at time t is
$$V_t^{(A)} = C_t + Ke^{-r(T-t)}.$$
At maturity T, the cash investment grows to K, hence
$$V_T^{(A)} = (S_T-K)^+ + K = \max(S_T,K).$$

- Portfolio B contains one European put option and one share of the underlying asset. Its value at time t is
$$V_t^{(B)} = P_t + S_t.$$
At maturity,
$$V_T^{(B)} = (K-S_T)^+ + S_T = \max(S_T,K).$$

We conclude that

$$V_T^{(A)} = V_T^{(B)}.$$

Since the two portfolios have identical value at maturity, the absence of arbitrage implies that their values at time t must also be equal:

$$C_t + Ke^{-r(T-t)} = P_t + S_t.$$

$$\iff C_t - P_t = S_t - Ke^{-r(T-t)}.$$